

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

CLICK PLUS Ladder Logic - Programming Reference

Powermatic 1200 drill press controls retrofit - CPU C2-01CPU-2 + 2x C2-14D1 - supersedes DWG 05 (Rev H large-format)

Rev J changes (2026-08-05)

1. X013 is the BH5928 immediate trip monitor through CR-S (1 = tripped). It removes C20, exits jog, and drives red immediately during the controlled-braking interval. X011 remains the delayed KA proof (1 = energized).
2. RUNG 1 and RUNG 23A now require X011 AND NOT X013. RUNG 19 cannot arm jog while X013 is active.
3. RUNG 21 exits jog on X013 immediately or on delayed loss of X011. RUNG 28/29 annunciation follows the same two-stage state: immediate red on X013, final-safe red on NOT X011.
4. No PLC rung initiates the E-stop. Two hardwired BH5928 N.O. contacts open DI1/DI2 immediately; delayed safety contacts subsequently open KA, ST01, and ST02.

Symbols

-] [- N.O. -]/[- N.C. -]UP[- press (leading edge) -]DN[- release (trailing edge)
-()- OUT -(S)- SET -(R)- RESET [TMR name Kvalue] timer NOT = NOT (logic annotations)

Internal bits & timers (Rev J)

C1 State IDLE (derived: TAP mode AND NOT C2 AND NOT C3 AND NOT fault)
C2 State TAP_DOWN (latched)
C3 State BACK_OUT (latched)
C10 FAULT - latched hard fault (manual clear via RUNG 17)
C11 THERMAL - DB over-temp, self-clearing (follows X012 with cool-down); inhibits run, blink-red
C20 RUN_PERMIT = X011 AND NOT X013 AND (X008 OR X009) AND X004 AND NOT C10 AND NOT C11
C30 DRILL_RUN - forward latch
C31 DRILL_REV - reverse latch
C40 Chunk-pulse request
C50 FWD-press-consumed flag - set by mode-entry hold (R19) or FWD-press-stops-REV (R7); while set, the press can neither release-latch a run (R4) nor drive jog motion (R24); clears on release (R8)
C60 JOG_MODE - mode armed (annunciated: green blink)
C61 JOG_RUN_OK - armed AND drum-ON AND STOP-clear AND X011 AND NOT X013 AND no fault/thermal. Arming does not require the drum; motion does.
C99 ~1 Hz free-running blink bit (indicator modulation)
T1 Chunk pulse timer (300 ms) T5FWD FWD hold timer (5 s -> arm jog)
T5STP STOP hold timer (5 s, jog exit) T60 Jog inactivity timer (60 s -> auto-exit)
Tcool DB cool-down delay (10 s) Tb1/Tb2 Blink astable (500/500 ms -> C99)

Scan order is load-bearing - transcribe rungs in the numbered order (23A between 23 and 24).

Rules preserved: (a) DRILL latch-SET rungs (4, 5) run before the opposite-stop rung (7); (b) C50 is set (R7/R19) before the FWD release-latch (R4) and jog legs (R24) evaluate it, and cleared last (R8); (c) C61 (R23A) is computed before the run-output rungs (R24/25) consume it.

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

PERMISSIVES & HOUSEKEEPING

RUNG 1 - RUN PERMISSIVE -> C20 (KA proved AND no immediate trip AND drum ON AND STOP clear AND healthy)

```
X011      X013      X008      X004      C10      C11
--] [-----]/[-----+----] [-----+--] [-----]/[-----]/[-----] ( )-- C20   RUN_PERMIT
          |      X009      | STOP(nc=1) noFault      noTherm
          +---] [-----+
```

C20 = X011 AND NOT X013 AND (X008 OR X009) AND X004 AND NOT C10 AND NOT C11
< REV J: X013 removes the permit immediately; X011 separately proves delayed KA closure.

RUNG 2 - SPEED PRESET -> Y003 (drum LOW -> 30 Hz; jog forces LOW)

```
X008
--] [-----+-----] ( )-- Y003                                PRESET_LOW
C60   | (jog armed -> force 30 Hz)
--] [-----+

Y003 = X008 OR C60
```

RUNG 3 - BLINK ASTABLE -> C99 (~1 Hz; or substitute a CLICK built-in clock relay and delete this rung)

```
C99
--]/[-----[TMR Tb1 K500ms]
Tb1
--] [-----] (S)-- C99
C99
--] [-----[TMR Tb2 K500ms]
Tb2
--] [-----] (R)-- C99
```

DRILL - LATCHED, BIDIRECTIONAL (stop-first)

RUNG 4 - DRILL FWD - LATCH SET on RELEASE (short tap latches; NOT C50 = press not consumed; NOT C60 = not in jog)

```
X001      X003      C50      C60      C20      C10      C31
--] [-----]DN[-----]/[-----]/[-----] [-----]/[-----]/[---(S)-- C30   DRILL_RUN
          release   NOT consumed   NOT jog      permit   noFault   NOT REV

SET C30 when: X001 AND X003DN AND NOT C50 AND NOT C60 AND C20 AND NOT C10 AND NOT C31
```

RUNG 5 - DRILL REV - LATCH SET on PRESS (mirror of FWD; no long-hold role, so acts on press)

```
X001      X005      C60      C20      C10      C30
--] [-----]UP[-----]/[-----] [-----]/[-----]/[-----] (S)-- C31   DRILL_REV
          press     NOT jog      permit   noFault   NOT FWD

SET C31 when: X001 AND X005UP AND NOT C60 AND C20 AND NOT C10 AND NOT C30
```

RUNG 6 - DRILL - LATCH RESET (stop / not DRILL / lost permit / fault -> drop both)

```
X004
--]/[-----+-----+-----] (R)-- C30
X001   |      +-----] (R)-- C31
--]/[-----+ (NOT DRILL)
C20    | (NOT permit - incl. X011 safety, drum-OFF, over-temp)
--]/[-----+
C10    | (fault)
--] [-----+

RESET C30,C31 when: NOT X004 OR NOT X001 OR NOT C20 OR C10
```

RUNG 7 - OPPOSITE-AS-STOP (opposite button = stop only; runs AFTER the SET rungs - scan order matters)

```
X005      C30
--]UP[-----] [-----+-----] (R)-- C30   REV press stops a FWD run
          +-] (S)-- C50   ?and consume

X003      C31
--]UP[-----] [-----+-----] (R)-- C31   FWD press stops a REV run
          +-] (S)-- C50   ?consume, so its RELEASE can't latch FWD (R4)
```

(Why it works: R5's NOT C30 / R4's NOT C31 interlocks block the opposite latch on the SAME press because the running latch is still set when SET evaluates; R7 then drops it. A second press latches.)

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

RUNG 8 - CLEAR FWD-CONSUMED ON RELEASE

```
X003
--]/[------(R)-- C50
```

RUNG 9 - DRILL BOTTOM-LIMIT FAULT (FWD only; reverse is not a feed-down runaway)

```
X001      C30      X006
--] [-----] [-----]/[------(S)-- C10  FAULT

SET C10 when: X001 AND C30 AND NOT X006          (X006=0 -> at bottom or wire break)
```

TAP - SINGLE-TAP STATE MACHINE

RUNG 10 - TAP START (IDLE -> TAP_DOWN) entry gate: lever OFF (X010=1), not in jog

```
X002      X003      X010      C60      C20      C10      C2      C3
--] [-----]UP[-----] [-----]/[-----] [-----]/[-----]/[-----]/[---(S)-- C2  TAP_DOWN
      leverOFF NOT jog      permit noFault

SET C2 when: X002 AND X003UP AND X010 AND NOT C60 AND C20 AND NOT C10 AND NOT C2 AND NOT C3
```

RUNG 11 - TAP_DOWN -> BACK_OUT (bottom limit)

```
C2      X006
--] [-----]/[-----+------(R)-- C2
      +------(S)-- C3          BACK_OUT
```

RUNG 12 - BACK_OUT -> IDLE (top limit) [note C: also SET C2 here for continuous tapping]

```
C3      X007
--] [-----]/[------(R)-- C3
```

RUNG 13 - TAP ABORT (stop / not TAP / fault / lost-permit-or-safety)

```
X004
--]/[-----+-----+------(R)-- C2
      X002 | +------(R)-- C3
--]/[-----+ (NOT TAP)
      C10 | (fault)
--] [-----+
      C20 | (lost permit - incl. E-stop/safety via X011, drum-OFF, STOP)
--]/[-----+
```

RUNG 14 - TAP FEED-LEVER RUNTIME TRIP (lever ON during a cycle = tap-snap)

```
C2      X010
--] [-----+]/[------(S)-- C10  FAULT
      C3 | LeverON
--] [-----+
```

DB OVER-TEMP (self-clearing) & FAULT RESET

RUNG 15 - DB OVER-TEMP SET (X012=0 = over-temp OR wire break -> assert immediately)

```
X012
--]/[------(S)-- C11  THERMAL
```

RUNG 16 - DB COOL-DOWN -> SELF-CLEAR (clears only after X012 stays normal for the cool-down delay)

```
X012
--] [-----[TMR Tcool K10000ms]
      Tcool
--] [------(R)-- C11

(Self-clearing: no manual reset; C11 is NOT the operator-clearable C10, so it can't fight R17.)
```

RUNG 17 - FAULT RESET (STOP while fully stopped clears the hard-fault latch)

```
X004      C2      C3      C30      C31
--]/[-----]/[-----]/[-----]/[-----]/[------(R)-- C10          FAULT
```

JOG MODE (Rev I behavior retained): hold-FWD is a MODE button - the entry hold never drives the spindle;

drum OFF permits mode switching but inhibits jog MOTION (see C61, RUNG 23A).

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

RUNG 18 - FWD HOLD TIMER (times how long FWD is held)

```
X003
--] [-----[TMR T5FWD K5000ms]
```

RUNG 19 - ENTER JOG MODE (FWD held >=5 s, safe, not already armed; consume the press so it can

```
neither latch a run on release (R4) nor jog while still held (R24))
T5FWD      C50      C60      X011      X013      C10
--] [-----]/[-----]/[-----] [-----]/[-----]/[-----+-(S)-- C60    JOG_MODE (armed -> green blink)
+-(S)-- C50    (entry press consumed)
```

Arming is allowed with the drum OFF - motion is not (C61, R23A).

RUNG 20 - JOG IDLE TIMEOUT (60 s with no jog button -> auto-exit)

```
C60      X003      X005
--] [-----]/[-----]/[-----[TMR T60 K60000ms]
T60
--] [------(R)-- C60
```

RUNG 21 - JOG FORCE-EXIT (any latched fault or safety trip)

```
C10
--] [-----+------(R)-- C60
X011 |
--]/[-----+ (delayed KA proof lost)
X013 |
--] [-----+ (immediate safety trip)
```

RUNG 22 - STOP HOLD TIMER (jog)

```
C60      X004
--] [-----]/[-----[TMR          T5STP      K5000ms]
```

RUNG 23 - SHORT-STOP EXIT (jog) (STOP released before 5 s -> exit; long hold = plain stop, stays armed)

```
C60      X004      T5STP
--] [-----]UP[-----]/[------(R)-- C60
STOP released NOT (was long hold)
```

RUNG 23A - JOG RUN PERMISSIVE -> C61 (armed AND drum ON AND STOP clear AND both safety indications healthy)

```
C60      X008      X004      X011      X013      C10      C11
--] [-----+--] [-----+--] [-----] [-----]/[-----]/[------( )-- C61    JOG_RUN_OK
| X009 |
+--] [-----+ (drum LOW OR HIGH)
```

C61 = C60 AND (X008 OR X009) AND X004 AND X011 AND NOT X013 AND NOT C10 AND NOT C11
< REV J: X013 blocks motion immediately during the controlled-braking interval.

VFD RUN OUTPUTS

RUNG 24 - VFD RUN-FWD -> Y001

```
C30      C20      Y002
--] [-----+--] [-----]/[------( )-- Y001    VFD_FWD
C2      | (normal: DRILL_RUN or TAP_DOWN)
--] [-----+
C61      | X003      C50      Y002
--] [-----+--] [-----]/[-----]/[--- (jog-FWD: hold-to-run; NOT C50 = the 5 s mode-entry hold can NOT jog)

Y001 = [ (C30 OR C2) AND C20 AND NOT Y002 ] OR [ C61 AND X003 AND NOT C50 AND NOT Y002 ]
Jog leg gated by C61 (adds drum/safety permissives) and by NOT C50 - in Rev H the spindle began jogging
at the 5 s mark while the entry press was still held. The mode-entry hold now drives nothing.
```

RUNG 25 - VFD RUN-REV -> Y002

```
C31      C20      Y001
--] [-----+--] [-----]/[------( )-- Y002    VFD_REV
C3      | (normal: DRILL_REV or TAP back-out)
--] [-----+
C61      | X005      Y001
--] [-----+--] [-----]/[--- (jog-REV: hold-to-run; REV has no mode-entry role, so no C50 term)

Y002 = [ (C31 OR C3) AND C20 AND NOT Y001 ] OR [ C61 AND X005 AND NOT Y001 ]
Jog leg gated by C61 (adds drum/safety permissives).
```

POWERMATIC 1200 RETROFIT - CLICK PLUS LADDER LOGIC

CHUNK (Contactor B - theater)

RUNG 26 - CHUNK REQUEST (edge of entering any run / direction - TAP or DRILL)

```

C2
--]UP[-----+------(S)-- C40   CHUNK_REQ
C3      |
--]UP[-----+
C30      |
--]UP[-----+
C31      |
--]UP[-----+
```

RUNG 27 - CHUNK TIMER + OUTPUT

```

C40
--] [-----[TMR T1 K300ms]
C40      T1
--] [-----]/[------( )-- Y004           CONTACTOR_B
T1
--] [------(R)-- C40
```

INDICATORS (4 states, exactly one active)

RUNG 28 - GREEN -> Y005 (solid = all-good; blink = jog armed)

```

X011      X013      C10      C11      C60
--] [-----]/[-----]/[-----]/[------( )-- Y005           GREEN
X011      | X013      C10      C11      C60      C99
--] [-----]/[-----]/[-----]/[-----] [-----] [--- (jog: blink via C99)

Y005 = X011 AND NOT X013 AND NOT C10 AND NOT C11 AND [ NOT C60 OR (C60 AND C99) ]
```

RUNG 29 - RED -> Y006 (solid = safety / hard fault; blink = over-temp)

```

X011
--]/[-----+------( )-- Y006   RED
X013      |
--] [-----+ (immediate trip)
C10      | (NOT KA proof OR immediate trip OR fault -> solid)
--] [-----+
C11      | X011      X013      C10      C99
--] [-----+ [-----]/[-----]/[-----] [--- (over-temp blink only when no solid-red)

Y006 = [ NOT X011 ] OR X013 OR C10 OR [ C11 AND X011 AND NOT X013 AND NOT C10 AND C99 ]
```

LADDER NOTES

- A - Force LOW in TAP (optional). R2 is Y003 = X008 OR C60. To also force LOW in every TAP cycle, add OR C2 OR C3. Left off so the drum is honored in TAP.
- B - FWD/REV interlock. R24/R25 each carry the other output's N.C.; the drive ramps through zero on reversal, so no PLC dead-time is required. Direction changes in DRILL are stop-first (R5/R6/R7), so the interlock only guards simultaneous-edge corner cases.
- C - Continuous tapping. In R12 also SET C2 when it RST C3 (top -> back down). One rung.
- D - First scan. IDLE is derived and every latch (C2 C3 C10 C11 C30 C31 C60) comes up cleared, so the machine powers up stopped, out of jog, and valid - no init rung.
- E - Max-dwell watchdog (optional). Time C2/C3; if a state overruns its limit, SET C10. Backstops the VFD over-torque trip and E-stop; does not replace them.
- F - C50 is the shared "this FWD press is spent" flag (Rev I: now triple-duty). Set when a 5 s hold arms jog (R19) or when a FWD press stops a running REV (R7). While set, the press can neither latch a forward run on release (R4) nor drive jog motion (R24). Clears on FWD release (R8). Scan order matters: R8 runs AFTER R4 and R24, so the flag is consumed before it clears.
- G - Blink source. R3 is a portable 1 Hz astable; CLICK PLUS also exposes a built-in clock-pulse relay you can assign to C99 and delete R3.
- H - Jog is advisory. Nothing in the jog rungs initiates the safety stop. X013 force-exits jog immediately; delayed loss of X011 separately proves KA has opened.
- I - Why exit is via STOP, not a second 5 s FWD hold (Rev I). In jog, holding FWD is hold-to-run - a legitimate jog can exceed 5 s, so a hold-again-to-exit would kick you out of the mode mid-jog. Exit paths: short (<5 s) STOP press - 60 s idle - any latched fault - E-stop/safety trip.

REV J VALIDATION DELTAS (add to section 10 checklist)

- [] With relay tripped but before KA opens: X013=1 and X011 may still be 1; C20, C60, and green all turn OFF immediately, red turns solid, and every run output/state clears.
- [] At delay expiry: KA opens and X011 becomes 0. Reset does not restore X011 until the BH5928 delay is complete, EDM is healthy, and a deliberate reset re-energizes KA.
- [] Open only the X013 field wire: final safe-state protection remains hardwired and X011 still falls when KA opens; diagnose the lost immediate advisory indication before returning to service.
- [] Hold FWD >=5 s with drum ON: green goes to blink at 5 s and the spindle does NOT turn while the entry press is still held; release, then a fresh FWD press jogs. (Verifies NOT C50 in R24.)
- [] With drum OFF: hold FWD >=5 s -> jog arms (green blink); FWD/REV presses do nothing; set drum to LOW -> presses now jog at 30 Hz. Return drum to OFF mid-hold -> motion stops. (Verifies C61.)
- [] Short STOP press exits jog; long STOP hold stops motion but stays armed; 60 s idle disarms.